Statistical Reasoning

Contents

3 Module 3 | The Big Picture

3.1 Module 3 | The Big Picture

In a nutshell, what statistics is all about is *converting data into useful information.* Statistics is therefore a process in which we

- Collect data,
- Summarize data, and

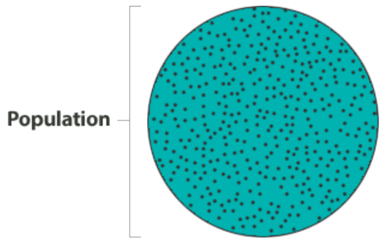
- Interpret data.

To really understand how this process works, we need to put it in a context. We will do that by introducing one of the central ideas of this course – the *Big Picture of Statistics*. We will introduce the Big Picture by building it gradually and explaining each step. At the end of the introductory explanation, once you have the full Big Picture in front of you, we will show it again using a concrete example.

The process of statistics starts when we identify what group we want to study or learn something about. We call this group the *population*. Note that the word *population* here (and in the entire course) does not refer only to people; it is used in the broader statistical sense to refer not only to people, but also to animals, objects, and so on. For example, we might be interested in

- The opinions of the population of U.S. adults about the death penalty
- How the population of mice react to a certain chemical
- The average price of the population of all one-bedroom apartments in a certain city

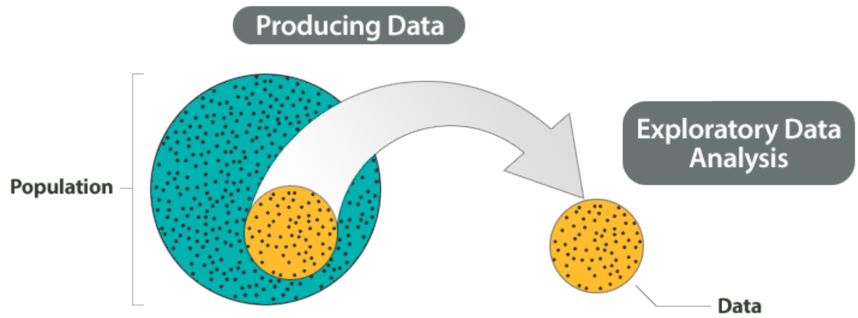
Population, then, is the entire group that is the target of our interest:



In most cases, the population is so large that, as much as we want to, there is absolutely no way we can study all of it (imagine trying to get the opinions of *all* U.S. adults about the death penalty). A more practical approach would be to examine and collect data only from a subgroup of the population, which we call a *sample*. We call this first step, which involves choosing a sample and collecting data from it, *producing data*.

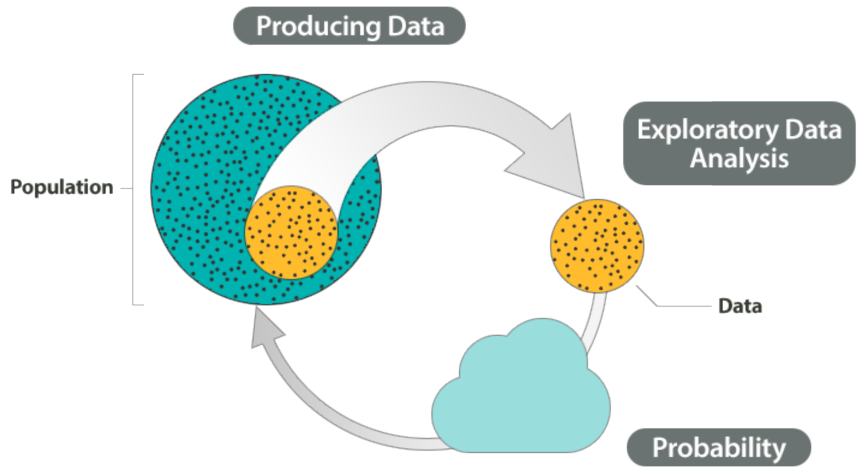
It should be noted that since, for practical reasons, we need to compromise and examine only a sub-group of the population rather than the whole population, we should make an effort to choose a sample in such a way that it will represent the population well. For example, if we choose a sample from the population of U.S. adults, and ask their opinions about the death penalty, we do not want our sample to consist of only Republicans or only Democrats.

Once the data have been collected, what we have is a long list of answers to questions, or numbers, and in order to explore and make sense of the data, we need to summarize that list in a meaningful way. This second step, which consists of summarizing the collected data, is called *exploratory data analysis*.

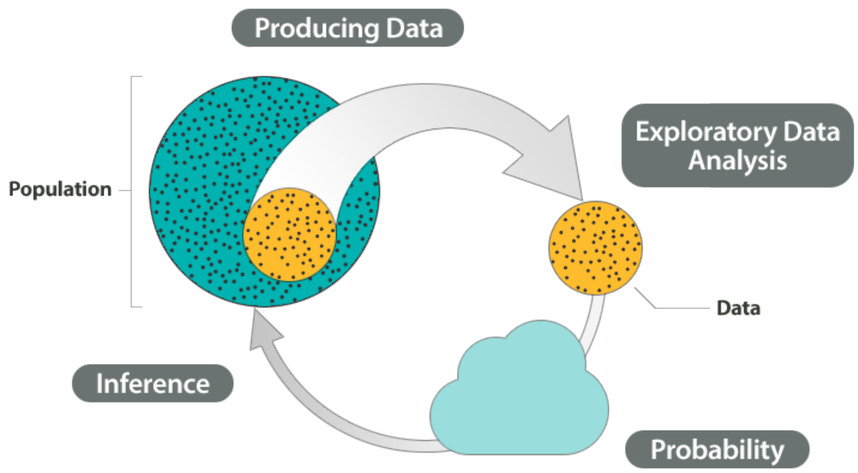


Now we've obtained the sample results and summarized them, but we are not done. Remember that our goal is to study the population, so what we want is to be able to draw conclusions about the population based on the sample results. Before we can do so, we need to look at how the sample we're using may differ from the population as a whole, so that we can factor that into our analysis. To examine this difference, we use *probability*.

In essence, probability is the "machinery" that allows us to draw conclusions about the population based on the data collected about the sample.



Finally, we can use what we've discovered about our sample to draw conclusions about our population. We call this final step in the process *inference*.



This is the *Big Picture* of statistics.

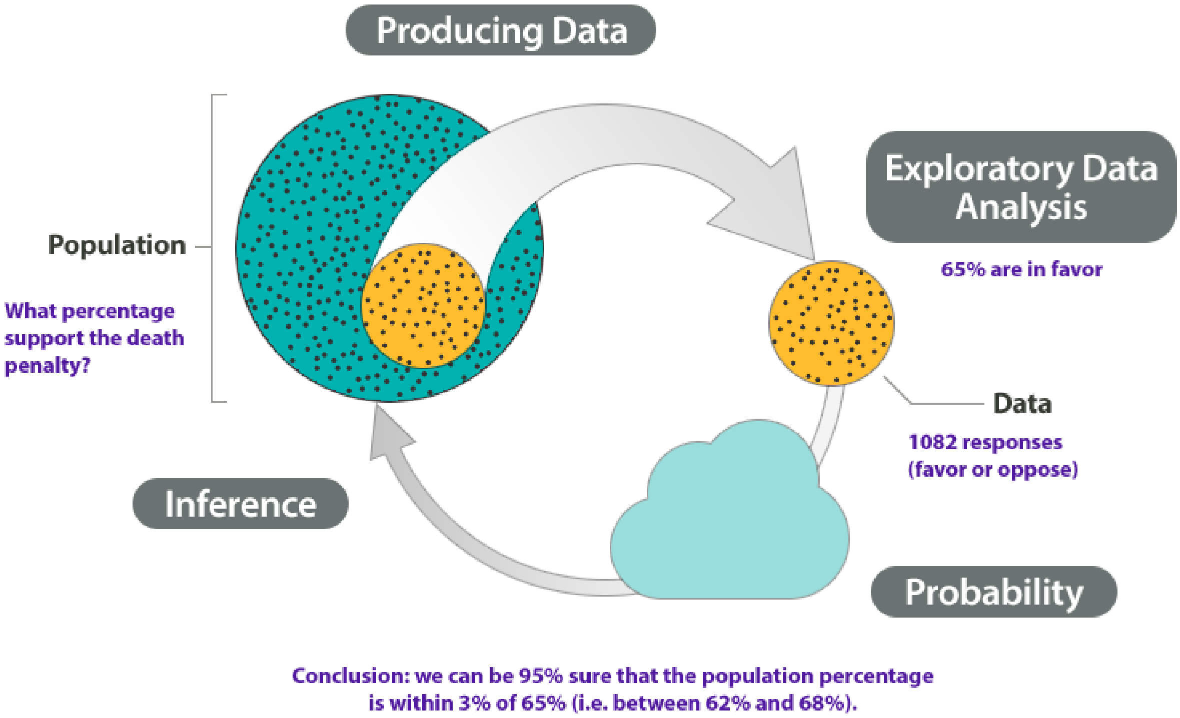
Example (A)

At the end of April 2005, a poll was conducted (by ABC News and the *Washington Post*) for the purpose of learning the opinions of U.S. adults about the death penalty.

1. *Producing Data:* A (representative) sample of 1,082 U.S. adults was chosen, and each adult was asked whether he or she favored or opposed the death penalty.

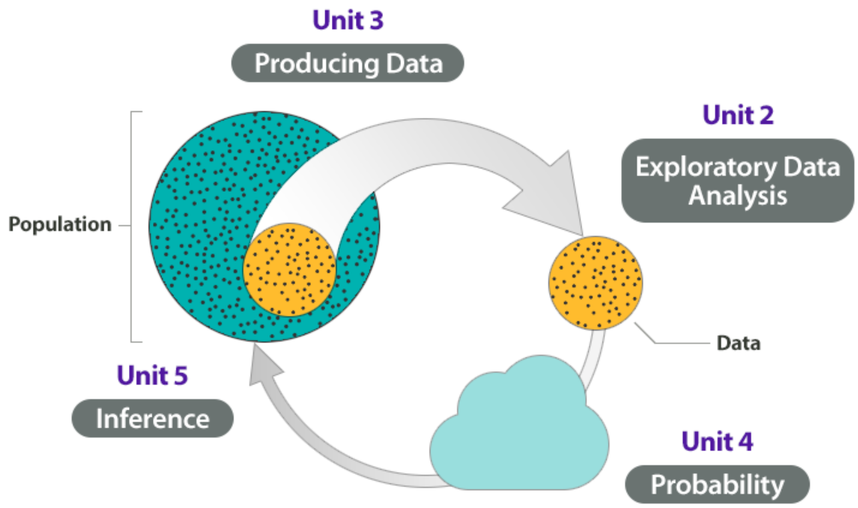
2. *Exploratory Data Analysis (EDA):* The collected data were summarized, and it was found that 65% of the sampled adults favor the death penalty for persons convicted of murder.

3 and 4. *Probability and Inference:* Based on the sample result (of 65% favoring the death penalty) and our knowledge of probability, it was concluded (with 95% confidence) that the percentage of those who favor the death penalty in the population is within 3% of what was obtained in the sample (i.e., between 62% and 68%). The following figure summarizes the example:



Structure The structure of this part of the course is based on the Big Picture. We start this course with EDA (even though it is second in the process of statistics), continue to discuss producing data, then proceed to probability, so that at the end we'll be able to discuss inference[1]. The following figure summarizes the structure of the course.

[1] KNOWLEDGE OF INFERENCE IS NOT A *CORE 200* LEARNING OUTCOME. WE SHALL NOT COVER IT



As you'll see, the Big Picture is the basis upon which the entire course is built, both conceptually and structurally. We will refer to it often, and having it in mind will help you as you go through the course.

4 Module 4 | Examining Distributions

4.1 Introduction

4.1.1 Exploratory Data Analysis (1 of 2)

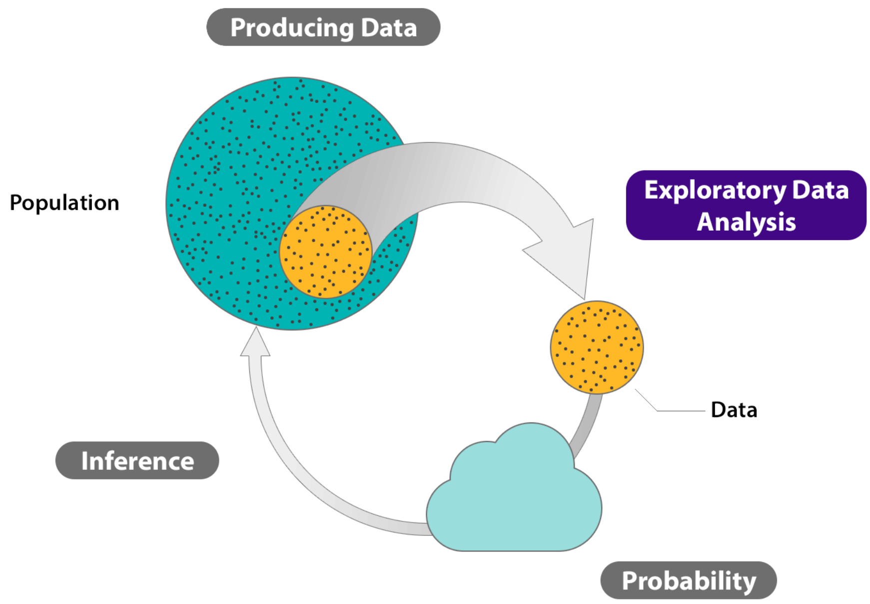
Recall the Big Picture, the four-step process that encompasses statistics (as it is presented in this course):

1. Producing Data – Choosing a sample from the population of interest and collecting data.

2. Exploratory Data Analysis (EDA) – Summarizing the data we've collected.

3. and 4. Probability and Inference – Drawing conclusions about the entire population based on the data collected from the sample.

Even though in practice it is the second step in the process, we look at exploratory data analysis (EDA) first.



4.1.2 Exploratory Data Analysis (2 of 2)

Before we jump into exploratory data analysis and really appreciate its importance in the process of statistical analysis, let's step back for a minute and ask:

What do we really mean by *data*?

Data are pieces of information about individuals organized into variables. By an *individual*, we mean a particular person or object. By a *variable*, we mean a particular characteristic of the individual.

A *dataset* is a set of data identified with particular circumstances. Datasets are typically displayed in tables, in which rows represent individuals and columns represent variables.

Example **(B) Medical Records**

The following dataset shows medical records from a particular survey:

Variables

Individuals

	Gender (M/F)	Age	Weight (lbs.)	Height (in.)	Smoking (1=No, 2=Yes)	Race
Patient #1	M	59	175	69	1	White
Patient #2	F	67	140	62	2	Black
Patient #3	F	73	155	59	1	Asian
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
Patient #75	M	48	90	72	1	White

In this example, the individuals are patients, and the variables are Gender, Age, Weight, Height, Smoking, and Race. Each row, then, gives us all the information about a particular individual (in this case, patient), and each column gives us information about a particular characteristic of all the patients.

Variables can be classified into one of two types: categorical or quantitative.

- *Categorical variables* take category or label values and place an individual into one of several groups. Each observation can be placed in *only* one category, and the categories are mutually exclusive.

  In our example of medical records, Smoking is a categorical variable, with two groups, since each participant can be categorized only as either a nonsmoker or a smoker. Gender and Race are the two other categorical variables in our medical records example. Notice that the values of the categorical variable Smoking have been coded as the numbers 1 or 2. It is common to code the values of a categorical variable as numbers, but you should remember that these are just codes. They have no arithmetic meaning (i.e., it does not make sense to add, subtract, multiply, divide, or compare the magnitude of such values).

- *Quantitative variables* take numerical values and represent some kind of measurement.

  In our medical example, Age is an example of a quantitative variable because it can take on multiple numerical values. It also makes sense to think about it in numerical form; that is, a person can be 18 years old or 80 years old. Weight and Height are also examples of quantitative variables.

Categorical variables are sometimes called *qualitative* variables, but in this course we use the term categorical.

4.1.3 A Sample Dataset

Background to Dataset Clinical depression is the most common mental illness in the United States, affecting 19 million adults each year (Source: NIMH, 1999). Nearly 50% of individuals who experience a major episode will have a recurrence within 2 to 3 years. Researchers are interested in comparing therapeutic solutions that could delay or reduce the incidence of recurrence.

In a study conducted by the National Institutes of Health, 109 clinically depressed patients were separated into three groups, and each group was given one of two active drugs (imipramine or lithium) or no drug at all. For each patient, the dataset contains the treatment used, the outcome of the treatment, and several other interesting characteristics.

Here is a summary of the variables in our dataset:

- *Hospt:* The patient's hospital, represented by a code for each of the 5 hospitals (1, 2, 3, 5, or 6)
- *Treat:* The treatment received by the patient (Lithium, Imipramine, or Placebo)
- *Outcome:* Whether or not a recurrence occurred during the patient's treatment (Recurrence or No Recurrence)
- *Time:* Either the time in days till the first recurrence, or if a recurrence did not occur, the length in days of the patient's participation in the study
- *AcuteT:* The time in days that the patient was depressed prior to the study
- *Age:* The age of the patient in years, when the patient entered the study
- *Gender:* The patient's gender (1 = Female, 2 = Male)

The actual dataset is printed below.

Dataset for Clinical Depression (Source: NIMH, 1999)

Hospt	Treat	Outcome	Time	AcuteT	Age	Gender
1	Lithium	Recurrence	36.143	211	33	1
1	Imipramine	No Recurrence	105.143	176	49	1
1	Imipramine	No Recurrence	74.571	191	50	1
1	Lithium	Recurrence	49.714	206	29	2
1	Lithium	No Recurrence	14.429	63	29	1
1	Placebo	Recurrence	5	70	30	2
1	Lithium	No Recurrence	104.857	55	56	1
1	Placebo	Recurrence	2.857	512	48	1
1	Placebo	No Recurrence	102.429	162	22	2
1	Placebo	Recurrence	55.714	306	61	2
1	Imipramine	No Recurrence	106.429	165	58	1
1	Imipramine	No Recurrence	105.143	129	31	1
1	Imipramine	No Recurrence	83	428	44	1
1	Imipramine	Recurrence	27.286	256	55	2
1	Lithium	No Recurrence	105.857	197	57	2
1	Lithium	Recurrence	5.571	227	46	1
1	Imipramine	No Recurrence	98	168	58	1
1	Lithium	No Recurrence	16.286	194	57	1
2	Lithium	Recurrence	1.286	173	54	1
2	Lithium	No Recurrence	2.143	48	23	1
2	Imipramine	No Recurrence	100	47	65	1
2	Imipramine	Recurrence	27.143	95	27	1
2	Lithium	Recurrence	4	148	50	1
2	Lithium	Recurrence	74.143	127	41	2

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Dataset for Clinical Depression (Source: NIMH, 1999) (Continued)

Hospt	Treat	Outcome	Time	AcuteT	Age	Gender
2	Placebo	No Recurrence	104.857	129	65	1
2	Placebo	Recurrence	0.143	182	52	1
2	Placebo	Recurrence	1.429	90	60	1
2	Placebo	Recurrence	45.857	177	25	2
2	Imipramine	Recurrence	17.429	234	27	2
2	Imipramine	No Recurrence	78	322	32	1
2	Imipramine	Recurrence	66.857	141	43	2
2	Placebo	No Recurrence	78.429	165	20	2
2	Lithium	No Recurrence	78.429	239	23	2
2	Imipramine	No Recurrence	78.143	147	36	2
2	Imipramine	No Recurrence	15.857	348	22	2
3	Lithium	No Recurrence	79	274	49	2
3	Imipramine	No Recurrence	32.571	130	40	2
3	Lithium	Recurrence	9	98	54	2
3	Lithium	Recurrence	3.286	77	26	1
3	Imipramine	No Recurrence	206	90	48	1
3	Lithium	Recurrence	30	280	51	2
3	Placebo	Recurrence	7.143	167	35	2
3	Placebo	Recurrence	31	181	28	1
3	Placebo	Recurrence	17.286	399	23	1
3	Placebo	Recurrence	0.143	289	57	2
5	Lithium	Recurrence	3.286	182	47	1
5	Imipramine	No Recurrence	1.571	159	31	2
5	Lithium	Recurrence	19.714	122	27	1
5	Imipramine	No Recurrence	126.714	115	61	1
5	Placebo	Recurrence	8	343	60	1
5	Lithium	Recurrence	71.714	114	28	1
5	Placebo	Recurrence	63.714	249	36	1
5	Placebo	No Recurrence	96.286	140	29	1
5	Lithium	No Recurrence	50.857	110	34	1
5	Imipramine	No Recurrence	155	214	49	1
5	Imipramine	No Recurrence	39.571	224	45	1
5	Lithium	Recurrence	36.286	294	28	1
5	Placebo	No Recurrence	102.571	162	24	2
5	Placebo	Recurrence	8.143	140	33	2
5	Imipramine	No Recurrence	28	147	34	1
5	Imipramine	No Recurrence	38	138	60	1
5	Imipramine	No Recurrence	111.571	196	23	2
5	Lithium	No Recurrence	165	139	35	1
5	Placebo	Recurrence	16	246	45	1
5	Lithium	No Recurrence	124.571	105	46	1
5	Lithium	No Recurrence	68	160	38	2

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Dataset for Clinical Depression (Source: NIMH, 1999) (Continued)

Hospt	Treat	Outcome	Time	AcuteT	Age	Gender
5	Placebo	No Recurrence	39.571	146	32	2
5	Placebo	No Recurrence	131	187	33	1
5	Imipramine	Recurrence	3.429	372	52	1
5	Lithium	No Recurrence	42	146	50	2
5	Imipramine	No Recurrence	26	131	38	1
5	Lithium	Recurrence	37.857	237	47	1
5	Imipramine	Recurrence	92.714	105	23	1
5	Imipramine	No Recurrence	106.714	140	31	1
5	Placebo	Recurrence	11.143	136	55	1
5	Placebo	No Recurrence	115	147	39	1
5	Placebo	Recurrence	44	160	41	1
5	Imipramine	No Recurrence	75	175	62	2
5	Placebo	No Recurrence	77.857	261	50	2
5	Lithium	Recurrence	0.286	146	46	1
5	Imipramine	No Recurrence	86	195	33	2
5	Placebo	No Recurrence	12.429	476	22	1
5	Lithium	No Recurrence	22	441	37	2
6	Lithium	Recurrence	5.429	86	40	2
6	Lithium	No Recurrence	67	201	22	1
6	Imipramine	Recurrence	3.429	130	30	2
6	Lithium	Recurrence	6.286	86	63	2
6	Imipramine	No Recurrence	5	209	40	1
6	Lithium	Recurrence	5.286	214	23	1
6	Imipramine	Recurrence	1	72	52	1
6	Placebo	Recurrence	3.429	238	23	1
6	Placebo	Recurrence	6.571	133	22	2
6	Placebo	Recurrence	1	128	23	1
6	Placebo	No Recurrence	45	139	30	2
6	Imipramine	No Recurrence	109.571	148	26	2
6	Lithium	Recurrence	0.857	285	46	1
6	Placebo	Recurrence	4.714	141	61	1
6	Imipramine	Recurrence	0.571	212	30	2
6	Imipramine	No Recurrence	9.143	168	39	1
6	Imipramine	No Recurrence	102	305	49	1
6	Lithium	Recurrence	46.286	204	57	1
6	Lithium	Recurrence	0.571	140	51	1
6	Lithium	Recurrence	6.429	182	53	1
6	Placebo	Recurrence	0	162	31	1
6	Placebo	Recurrence	20.857	207	43	1
6	Placebo	Recurrence	18.286	102	29	1
6	Imipramine	Recurrence	31.857	154	28	1
6	Imipramine	Recurrence	22	203	51	1

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Dataset for Clinical Depression (Source: NIMH, 1999) (Continued)

Hospt	Treat	Outcome	Time	AcuteT	Age	Gender
6	Lithium	Recurrence	2	176	33	1

4.1.4 Scales of Measurement / Types of Variables

In the previous section, a simple distinction was made between quantitative and categorical variables. However, there is a more precise method of categorizing variables: it is called *scale of measurement*. The four different scales of measurement, from least to most precise, are

- Nominal
- Ordinal
- Interval
- Ratio

In the walkthrough video at `https://www.youtube.com/watch?v=dbI8EDP8IxE` and the following pages, each one of these types of variables is described with a comparison of their properties.

Nominal Scale of Measurement The *nominal scale of measurement* is a qualitative measure that uses discrete categories to describe a characteristic of the research participants. For each participant, the researcher determines the presence, absence, and type of the attribute. Nominal scales of measurement may have two categories, such as citizen status (citizen/non-citizen), or they can have more than two categories, like religious affiliation (e.g., Agnostic, Buddhist, Jewish, Muslim) or marital status (e.g., divorced, married, single). Often, as described here, the categories have names; however, researchers code them with numbers for use in statistical analyses. These categories are not ordered or ranked in any way.

Ordinal Scale of Measurement An *ordinal scale of measurement* rank-orders participants on some scale or attribute, but the difference between numbers does not convey fixed or equal differences. Thus, with ordinal data, we know that a one-unit increase in an ordinal scales represents "more," but we don't know how much more. For example, a group of participants can be rank-ordered from least to most politically active. We know that a person who is ranked as 5 is more politically active than a person who is ranked as 4, but not how much more politically active. The value of the variable is used to order participants according to the strength/presence of the attribute and not to calculate differences between participants.

Interval Scale of Measurement The *interval scale of measurement* takes numerical form, and the distance between pairs of consecutive numbers is assumed to be equal. However, interval variables do not have a meaningful zero point; thus, a zero does not mean the absence of the attribute, but rather it is a particular (but arbitrary) point on the scale. A good example of an interval measure is temperature in the Fahrenheit scale: a temperature of zero degrees Fahrenheit is still a temperature, not the absence of temperature. In education, measures like achievement, motivation, and self-concept are considered interval measures; a zero on a measure of such variables does not mean the absence of the characteristic in the participant.

Ratio Scale of Measurement The *ratio scale of measurement* is similar to the interval scale. As with the interval scale, a number is assigned to a subject that represents the amount of the attribute that the subject has and the difference between consecutive numbers is assumed to be equal. The main difference between interval and ratio measurements has to do with how we interpret a value of zero. For *ratio* measures, the zero *is* meaningful and tell us that the attribute is not present in the participant. Examples of ratio measures include a participant's number of children, number of AP courses taken, or cumulative college credits: for each of these variables, a score of zero represents that the participant has none of the attribute.

Learning Objectives

- Summarize and describe the distribution of a categorical variable in context.
- Generate and interpret several different graphical displays of the distribution of a quantitative variable (histogram, stemplot, boxplot).
- Summarize and describe the distribution of a quantitative variable in context: a) describe the overall pattern, b) describe striking deviations from the pattern.
- Relate measures of center and spread to the shape of the distribution, and choose the appropriate measures in different contexts.
- Compare and contrast distributions (of quantitative data) from two or more groups, and produce a brief summary, interpreting your findings in context.
- Apply the standard deviation rule to the special case of distributions having the "normal" shape.

4.1.5 Module 4 | Examining Distributions

As indicated in the introduction, we will begin the EDA part of the course by exploring (or looking at) one variable at a time.

As we saw in the Introduction to Exploratory Data Analysis, the data for each variable are a long list of values (whether numerical or not), and are not very informative in that form. In order to convert these raw data into useful information we need to summarize and then examine the *distribution* of the variable. By *distribution* of a variable, we mean:

- what values the variable takes, and
- how often the variable takes those values.

This module has two sections. We will first learn how to summarize and examine the distribution of a single categorical variable, and then do the same for a quantitative variable.

4.2 One Categorical Variable

4.2.1 Module 4 | One Categorical Variable (1 of 3)

Learning Objectives

- Summarize and describe the distribution of a categorical variable in context.

What is your perception of your own body? Do you feel that you are overweight, underweight, or about right?

A random sample of 1,200 U.S. college students were asked this question as part of a larger survey. The following table shows part of the responses:

Student	Body Image
student 25	overweight
student 26	about right
student 27	underweight
student 28	about right
student 29	about right

Here is some information that would be interesting to get from these data:

- What percentage of the sampled students fall into each category?
- How are students divided across the three body image categories? Are they equally divided? If not, do the percentages follow some other kind of pattern?

There is no way that we can answer these questions by looking at the raw data, which are in the form of a long list of 1,200 responses, and thus not very useful. However, both these questions will be easily answered once we summarize and look at the *distribution* of the variable Body Image (i.e., once we summarize how often each of the categories occurs).

In order to summarize the distribution of a categorical variable, we first create a table of the different values (categories) the variable takes, how many times each value occurs (count) and, more importantly, how often each value occurs (by converting the counts to percentages); this table is called a frequency distribution. Here is the frequency distribution for our example:

Category	Count	Percent
About right	855	$\left(\frac{855}{1200}\right) \times 100 = 71.3\%$
Overweight	235	$\left(\frac{235}{1200}\right) \times 100 = 19.6\%$
Underweight	110	$\left(\frac{110}{1200}\right) \times 100 = 9.2\%$
Total	*n=1200*	*100%*

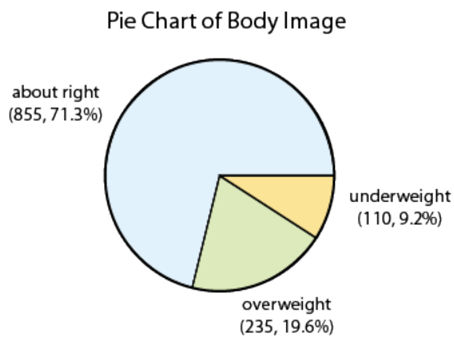
4.2.2 Module 4 | One Categorical Variable (2 of 3)

Learning Objectives

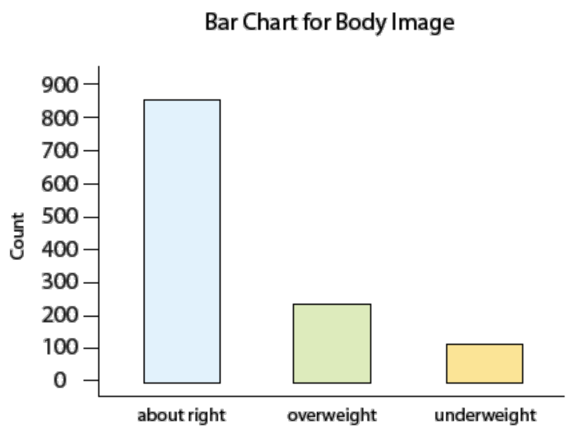
- Summarize and describe the distribution of a categorical variable in context.

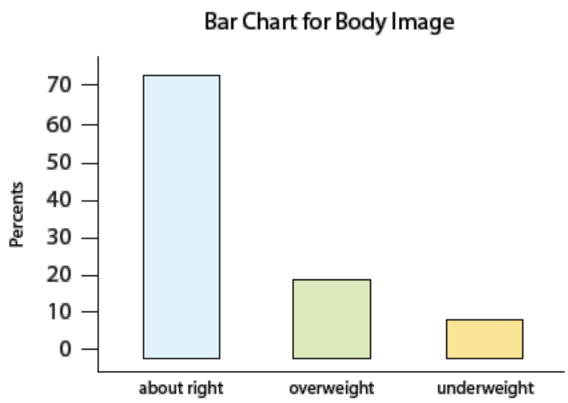
In order to visualize the numerical summaries we've obtained, we need a graphical display. There are two simple graphical displays for visualizing the distribution of categorical data:

1. The Pie Chart



2. The Bar Chart





Now that we have summarized the distribution of values in the Body Image variable, let's go back and interpret the results in the context of the questions that we posed:

Now that we've interpreted the results, there are some other interesting questions that arise:

- Can we reliably generalize our results to the entire population of interest and conclude that a similar distribution across body image categories exists among all U.S. college students? In particular, can we make such a generalization even though our sample consisted of only 1,200 students, which is a very small fraction of the entire population?
- If we had separated our sample by gender and looked at males and females separately, would we have found a similar distribution across body image categories?

These are the types of questions that we will deal with in future sections of the course.

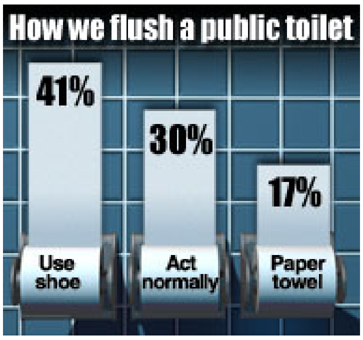
4.2.3 Module 4 | One Categorical Variable (3 of 3)

Learning Objectives

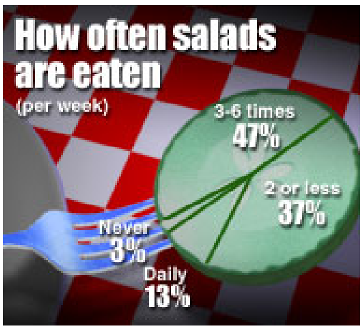
- Summarize and describe the distribution of a categorical variable in context.

Comments:

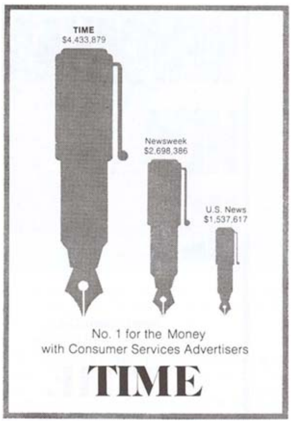
1. While both the pie chart and the bar chart help us visualize the distribution of a categorical variable, the pie chart emphasizes how the different categories relate to the whole, and the bar chart emphasizes how the different categories compare with each other.
2. A variation on the pie chart and bar chart that is very commonly used in the media is the *pictogram*. Here are two examples:

   Source: USA Today Snapshots and the Impulse Research for Northern Confidential Bathroom survey

   Source: Market Facts for the Association of Dressings and Sauces
3. *Beware:* Pictograms can be misleading. Consider the following pictogram:



This graph is aimed at advertisers deciding where to spend their budgets, and clearly suggests that *Time* magazine attracts by far the largest amount of advertising spending. Are the differences really as dramatic as the graph suggests? If we look carefully at the numbers above the pens, we find that advertisers spend in *Time* only $4,433,879 / $2,698,386 = 1.64 times more than in *Newsweek*, and only $4,433,879 / $1,537,617 = 2.88 times more than in *U.S. News*. By looking at the pictogram, however, we get the impression that *Time* is much further ahead. Why? In order to magnify the picture without distorting it, we must increase *both* its height and width. As a result, the *area* of *Time's* pen is 1.64 × 1.64 = 2.7 times larger than the *Newsweek* pen, and 2.88 × 2.88 = 8.3 times larger than the *U.S. News* pen. Our eyes capture the area of the pens rather than only the height, and so we are misled to think that *Time* is a bigger winner than it really is.

4.2.4 Let's Summarize

- The distribution of a categorical variable is summarized using:
  - *Graphical display:* pie chart or bar chart, supplemented by
  - *Numerical summaries:* category counts and percentages.
- A variation on pie charts and bar charts is the pictogram.
- Pictograms can be misleading, so make sure to use a critical approach when interpreting the information the pictogram is trying to convey.

4.3 Module 4 | One Quantitative Variable

4.3.1 Introduction

In the previous section, we explored the distribution of a categorical variable using graphs (pie chart, bar chart) supplemented by numerical measures (percent of observations in each category). In this section, we will explore the data collected from a *quantitative* variable, and learn how to describe and summarize the important features of its distribution. We will first learn how to display the distribution using graphs and then move on to discuss numerical measures.

4.3.2 Module 4 | Graphs

To display data from one quantitative variable graphically, we can use either the *histogram* or the *stemplot*. (Another graph, the *boxplot*, will be mentioned later in this module).

4.3.3 Module 4 | Histogram (1 of 3)

Learning Objectives

- Generate and interpret several different graphical displays of the distribution of a quantitative variable (histogram, stemplot, boxplot).

Idea Break the range of values into intervals and count how many observations fall into each interval.

Example (C) Exam Grades

Here are the exam grades of 15 students:

Score	Count
[40–50)	1
[50–60)	2
[60–70)	4
[70–80)	5
[80–90)	2
[90–100]	1

Note: The observation 60 was counted in the 60–70 interval. See comment 1 below.

To construct the histogram from this table we plot the intervals on the X-axis, and show the number of observations in each interval (frequency of the interval) on the Y-axis, which is represented by the height of a rectangle located above the interval:

The table above can also be turned into a relative frequency table using the following steps:

1. Add a row on the bottom and include the total number of observations in the dataset that are represented in the table.
2. Add a column, at the end of the table, and calculate the relative frequency for each interval, by dividing the number of observations in each row by the total number of observations.

These two steps are illustrated in red in the following frequency distribution table:

Score	Count (also called Frequency)	Relative Frequency
[40–50)	1	0.07
[50–60)	2	0.13
[60–70)	4	0.27
[70–80)	5	0.33
[80–90)	2	0.13
[90–100]	1	0.07
Total	15	

Step 1: Add a row at bottom of table. Put in total number of observations in the data set.

In this example, there are 15 (1+2+4+5+2+1= 15) total observations.

Step 2: Add a column to right side of table. Determine the relative frequencies of each interval by dividing the interval count (or frequency) by the total number of observations.

For example, to determine the relative frequency of scores in the [40-50) interval, divide the count (or frequency) by the total number of observations: 1 /15 = .07.

The relative frequency for the [50-60) interval is: 2/15 = .13.

Continue until all of the relative frequencies have been calculated.

To convert each relative frequency into a percentage, multiply it by 100. For example, the percentage of scores for the [40-50) interval would be .07*100= 7, which is 7%.

It is also possible to determine the number of scores for an interval, if you have the total number of observations and the relative frequency for that interval. For instance, suppose there are 15 scores (or observations) in a set of data and the relative frequency for an interval is 0.13. To determine the number of scores in that interval, multiplying the total number of observations by the relative frequency and round up to the next whole number: $15 \times 0.13 = 1.95$, which rounds up to 2 observations.

A relative frequency table, like the one above, can be used to determine the frequency of scores occurring at or across intervals. Here are some examples, using the above frequency table:

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1. What is the percentage of exam scores that were 70 and up to, but not including, 80? To determine the answer, we look at the relative frequency associated with the [70-80) interval. The relative frequency is 0.33; to convert to percentage, multiply by 100 (0.33 × 100= 33) or 33%.
2. What is the percentage of exam scores that are at least 70? To determine the answer, we need to:
   - Add together the relative frequencies for the intervals that have scores of at least 70 or above. Thus, would need to add together the relative frequencies from [70-80), [80-90), and [90-100] = 0.33+.13+.07 = 0.53.
   - To get the percentage, need to multiple the calculated relative frequency by 100. In this case, it would be 0.53 × 100 = 53 or 53%.

Comments:

1. It is very important that each observation be counted only in one interval. For the most part, it is clear which interval an observation falls in. However, in our example, we needed to decide whether to include 60 in the interval 50–60, or the interval 60–70, and we chose to count it in the latter. In fact, this decision is captured by the way we wrote the intervals. If you scroll up and look at the table, you'll see that we wrote the intervals in a peculiar way: [40–50), [50,60), [60,70), and so on. The square bracket means "including," and the parenthesis means "not including." For example, [50,60) is the interval from 50 to 60, including 50 and not including 60; [60,70) is the interval from 60 to 70, including 60, and not including 70. It does not matter how you decide to set up your intervals as long as you're consistent.

2. When data are displayed in a histogram, some information is lost. Note that by looking at the histogram we *can* answer: "How many students scored 70 or above?" (5 + 2 + 1 = 8) But we *cannot* answer: "What was the lowest score?" All we can say is that the lowest score is somewhere between 40 and 50, and therefore we can approximate that it is around 45.

3. Obviously, we could have chosen to break the data into intervals differently (for example, 45–50, 50–55, 55–60).

How do I know what interval width to choose? There are no right or wrong choices of interval widths. In this course, we will rely on a statistical package to produce the histogram for us, and we will focus instead on describing and summarizing the distribution as it appears from the histogram.

4.3.4 Module 4 | Histogram (2 of 3)

Learning Objectives

- Generate and interpret several different graphical displays of the distribution of a quantitative variable (histogram, stemplot, boxplot).
- Summarize and describe the distribution of a quantitative variable in context: a) describe the overall pattern, b) describe striking deviations from the pattern.

Interpreting the Histogram Once the distribution has been displayed graphically, we can describe the overall pattern of the distribution and mention any striking deviations from that pattern. More specifically, we should consider the following features of the distribution:

- Shape
- Center } overall pattern
- Spread
- Outliers → deviations from the pattern

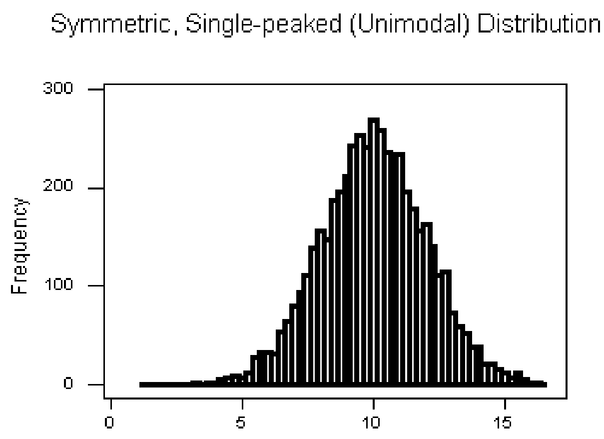
We will get a sense of the overall pattern of the data from the histogram's center, spread and shape, while outliers will highlight deviations from that pattern.

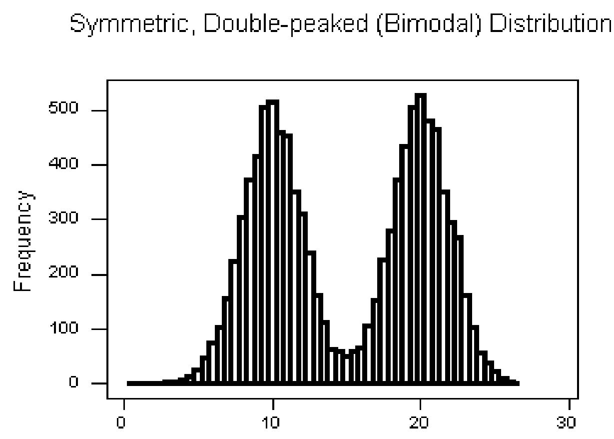
Shape When describing the shape of a distribution, we should consider:

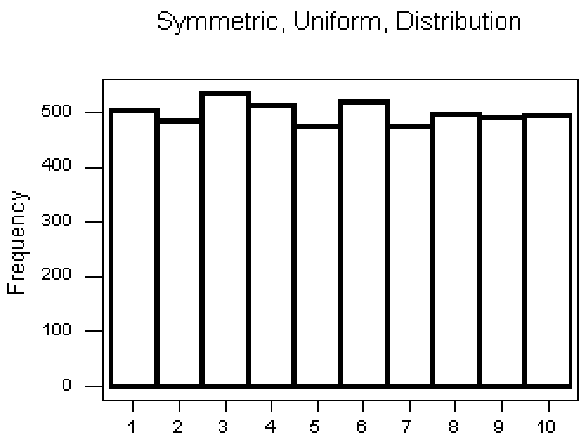
1. *Symmetry/skewness* of the distribution.
2. *Peakedness (modality)* – the number of peaks (modes) the distribution has.

We distinguish between:

Symmetric Distributions

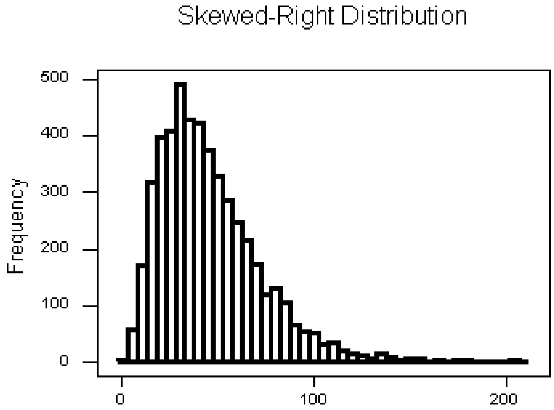






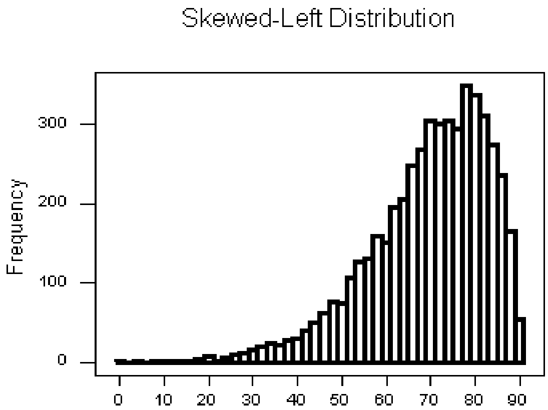
Note that all three distributions are symmetric, but are different in their modality (peakedness). The first distribution is *unimodal* – it has one mode (roughly at 10) around which the observations are concentrated. The second distribution is *bimodal* – it has two modes (roughly at 10 and 20) around which the observations are concentrated. The third distribution is kind of flat, or *uniform*. The distribution has no modes, or no value around which the observations are concentrated. Rather, we see that the observations are roughly uniformly distributed among the different values.

Skewed Right Distributions



A distribution is called *skewed right* if, as in the histogram above, the right tail (larger values) is much longer than the left tail (small values). Note that in a skewed right distribution, the bulk of the observations are small/medium, with a few observations that are much larger than the rest. An example of a real-life variable that has a skewed right distribution is salary. Most people earn in the low/medium range of salaries, with a few exceptions (CEOs, professional athletes etc.) that are distributed along a large range (long "tail") of higher values.

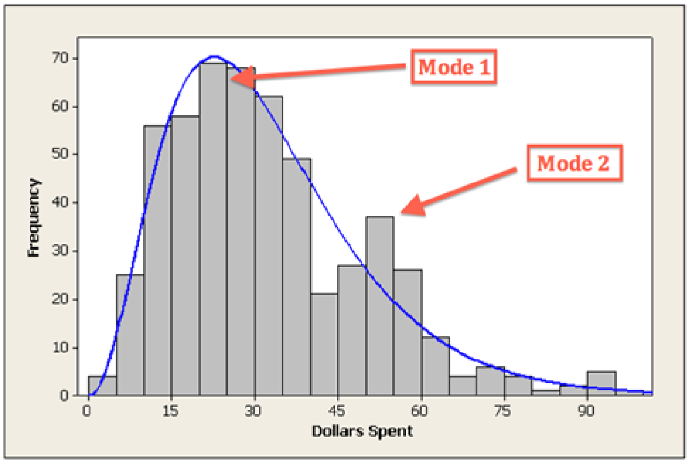
Skewed Left Distributions



A distribution is called *skewed left* if, as in the histogram above, the left tail (smaller values) is much longer than the right tail (larger values). Note that in a skewed left distribution, the bulk of the observations are medium/large, with a few observations that are much smaller than the rest. An example of a real life variable that has a skewed left distribution is age of death from natural causes (heart disease, cancer etc.). Most such deaths happen at older ages, with fewer cases happening at younger ages.

Comments:

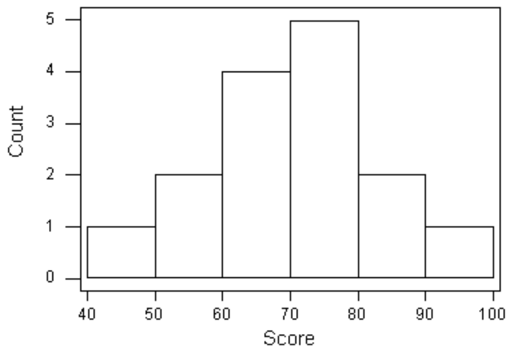
1. Note that skewed distributions can also be bimodal. Here is an example. A medium size neighborhood 24-hour convenience store collected data from 537 customers on the amount of money spend in a single visit to the store. The following histogram displays the data.



Note that the overall shape of the distribution is skewed to the right with a clear mode around \$25. In addition it has another (smaller) "peak" (mode) around \$50-55. The majority of the customers spend around \$25 but there is a cluster of customers who enter the store and spend around \$50-55.

2. If a distribution has more than two modes, we say that the distribution is *multimodal*.

Recall our grades example:



As you can see from the histogram, the grades distribution is roughly symmetric.

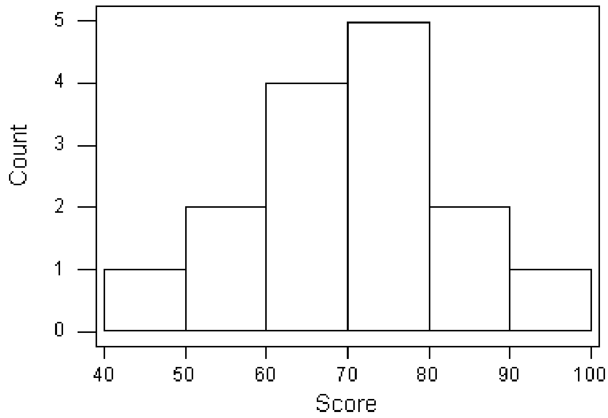
4.3.5 Module 4 | Histogram (3 of 3)

Learning Objectives

- Generate and interpret several different graphical displays of the distribution of a quantitative variable (histogram, stemplot, boxplot).
- Summarize and describe the distribution of a quantitative variable in context: a) describe the overall pattern, b) describe striking deviations from the pattern.

Center The center of the distribution is its *midpoint* – the value that divides the distribution so that approximately half the observations take smaller values, and approximately half the observations take larger values. Note that from looking at the histogram we can get only a rough estimate for the center of the distribution. (More exact ways of finding measures of center will be discussed in the next section.)

Recall our grades example:



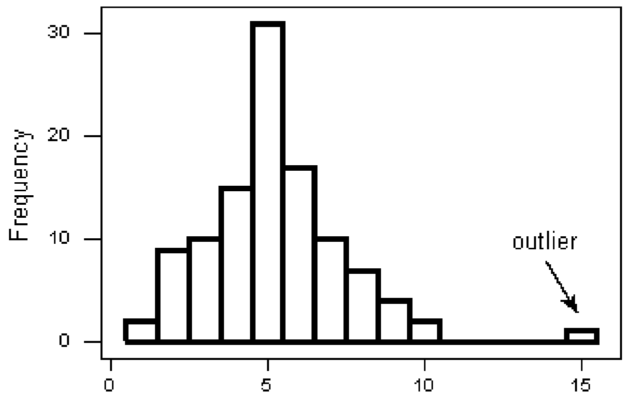
As you can see from the histogram, the center of the grades distribution is roughly 70 (7 students scored below 70, and 8 students scored above 70).

Spread The *spread* (also called *variability*) of the distribution can be described by the approximate range covered by the data. From looking at the histogram, we can approximate the smallest observation (*min*), and the largest observation (*max*), and thus approximate the range. (More exact ways of finding measures of spread are discussed in the next section.)

In our example:

Approximate min:	45 (the middle of the lowest interval of scores)
Approximate max:	95 (the middle of the highest interval of scores)
Approximate range:	95 – 45 = 50

Outliers *Outliers* are observations that fall outside the overall pattern. For example, the following histogram represents a distribution that has a high probable outlier:



Go back and check the histogram of scores at the top of this page. As you can see, there are no outliers.

Example (D) Best Actress Oscar Winners

To provide an example of a histogram applied to actual data, we will look at the ages of Best Actress Oscar winners from 1970 to 2013. The full dataset is tabulated below.

Best Actress Oscar Winners (1970–2013)

Year	Name	Movie	Age
1970	Glenda Jackson	Women in Love	34
1971	Jane Fonda	Klute	34
1972	Liza Minnelli	Cabaret	27
1973	Glenda Jackson	A Touch of Class	37
1974	Ellen Burstyn	Alice Doesn't Live Here Anymore	42
1975	Louise Fletcher	One Flew over the Cuckoo's Nest	41
1976	Faye Dunaway	Network	36
1977	Diane Keaton	Annie Hall	32
1978	Jane Fonda	Coming Home	41
1979	Sally Field	Norma Rae	33
1980	Sissy Spacek	Coal Miner's Daughter	31
1981	Katharine Hepburn	On Golden Pond	74
1982	Meryl Streep	Sophie's Choice	33
1983	Shirley MacLaine	Terms of Endearment	49
1984	Sally Field	Places in the Heart	38
1985	Geraldine Page	A Trip to the Bountiful	61
1986	Marlee Matlin	Children of a Lesser God	21
1987	Cher	Moonstruck	41
1988	Jodie Foster	The Accused	26
1989	Jessica Tandy	Driving Miss Daisy	80
1990	Kathy Bates	Misery	42
1991	Jodie Foster	The Silence of the Lambs	29
1992	Emma Thompson	Howards End	33
1993	Holly Hunter	The Piano	36
1994	Jessica Lange	Blue Sky	45
1995	Susan Sarandon	Dead Man Walking	49
1996	Frances McDormand	Fargo	39
1997	Helen Hunt	As Good As It Gets	34
1998	Gwyneth Paltrow	Shakespeare in Love	26
1999	Hilary Swank	Boys Don't Cry	25
2000	Julia Roberts	Erin Brockovich	33
2001	Halle Berry	Monster's Ball	35
2002	Nicole Kidman	The Hours	35
2003	Charlize Theron	Monster	28
2004	Hilary Swank	Million Dollar Baby	30
2005	Reese Witherspoon	Walk the Line	29
2006	Helen Mirren	The Queen	61

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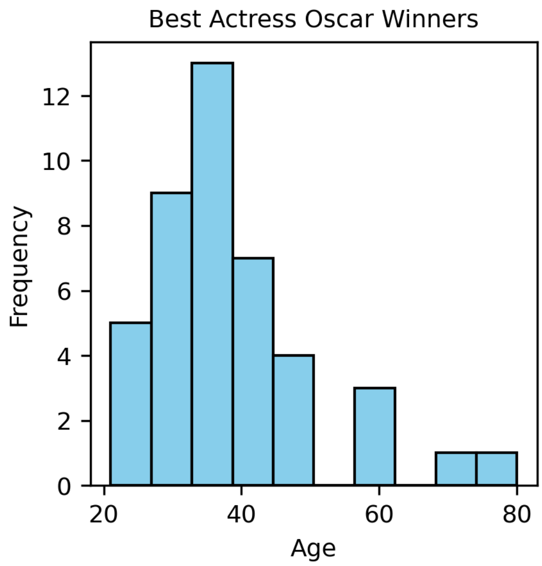
Best Actress Oscar Winners (1970–2013) (Continued)

Year	Name	Movie	Age
2007	Marion Cotillard	La Vie en Rose	32
2008	Kate Winslet	The Reader	33
2009	Sandra Bullock	The Blind Side	45
2010	Natalie Portman	Black Swan	29
2011	Meryl Streep	The Iron Lady	62
2012	Jennifer Lawrence	Silver Linings Playbook	22
2013	Cate Blanchett	Blue Jasmine	44

The histogram for the data is shown below.



We will now summarize the main features of the distribution of ages as it appears from the histogram:

Shape: The distribution of ages is skewed right. We have a concentration of data among the younger ages and a long tail to the right. The vast majority of the "best actress" awards are given to young actresses, with very few awards given to actresses who are older.

Center: The data seem to be centered around 34 or 35 years old. Note that this implies that roughly half the awards are given to actresses who are less than 34 years old.

Spread: The data range from about 20 to about 80, so the approximate range equals 80 - 20 = 60.

Outliers: There seem to be two probable outliers to the far right and possibly three around 62 years old.

You can see how informative it is to know "what to look at" in a histogram. If there is one conclusion that we can make here, it is that Hollywood likes to give Oscars to young actresses.

4.3.6 Let's Summarize

- The histogram is a graphical display of the distribution of a quantitative variable. It plots the number (count) of observations that fall in intervals of values.
- When examining the distribution of a quantitative variable, one should describe the overall pattern of the data (shape, center, spread), and any deviations from the pattern (outliers).
- When describing the shape of a distribution, one should consider:
  - Symmetry/skewness of the distribution

– Peakedness (modality) – the number of peaks (modes) the distribution has.

Not all distributions have a simple, recognizable shape.

- Outliers are data points that fall outside the overall pattern of the distribution and need further research before continuing the analysis.
- It is always important to interpret what the features of the distribution (as they appear in the histogram) mean in the context of the data.

4.3.7 Module 4 | Stemplot

Learning Objectives

- Summarize and describe the distribution of a quantitative variable in context: a) describe the overall pattern, b) describe striking deviations from the pattern.

The stemplot (also called stem and leaf plot) is another graphical display of the distribution of quantitative data.

Idea Separate each data point into a stem and leaf, as follows:

- The leaf is the right-most digit.
- The stem is everything except the right-most digit.
- So, if the data point is 34, then 3 is the stem and 4 is the leaf.
- If the data point is 3.41, then 3.4 is the stem and 1 is the leaf.

Example **(E) Best Actress Oscar Winners**

We will continue with the Best Actress Oscar winners example.

34 34 27 37 42 41 36 32 41 33 31 74 33 49 38 61 21 41 26 80 42 29 33 36 45 49 39 34 26 25 33 35 35 28 30 29 61 32 33 45 29 62 22 44

To make a stemplot:

1. Separate each observation into a stem and a leaf.
2. Write the stems in a vertical column with the smallest at the top, and draw a vertical line at the right of this column.
3. Go through the data points, and write each leaf in the row to the right of its stem.
4. Rearrange the leaves in an increasing order.

```
Steps 1, 2, and 3           Step 4

2|7169658992                2|1256678999                2|12
3|33762313836943550 23      3|0122333333445566789       2|56678999
4|2119124954                4|1112244599                3|012233333344
5|                  →       5|                  →       3|5566789
6|112                       6|112                       4|1112244
7|4                         7|4                         4|599
8|0                         8|0                         5|
                                                        5|
                                                        6|112
                                                        6|
                                                        7|4
                                                        7|
                                                        8|0
```

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When some of the stems hold a large number of leaves, we can split each stem into two: one holding the leaves 0-4, and the other holding the leaves 5-9. A statistical software package will often do the splitting for you, when appropriate.

Note that when rotated 90 degrees counterclockwise, the stemplot visually resembles a histogram:

```
    4
    4
    3
    3
  9 3
  9 3 9 4
  9 3 8 4
  8 3 7 2
  7 2 6 2
  6 2 6 1 9     2
2 6 1 5 1 9     1
1 5 0 5 1 5     1   4   0
-------------------------
2 2 3 3 4 4 5 5 6 6 7 7 8
```

This orientation makes the right-skewedness of the distribution clearly visible.

The stemplot has additional unique features:

- It preserves the original data.
- It sorts the data (which will become very useful in the next section).

Comment There is another type of display that we can use to summarize a quantitative variable graphically – the *dotplot*. The dotplot, like the stemplot, shows each observation, but displays it with a dot rather than with its actual value.

4.3.8 Let's Summarize

The stemplot is a simple but useful visual display of quantitative data. Its principal virtues are:

- Easy and quick to construct for small, simple datasets.
- Retains the actual data.
- Sorts (ranks) the data.

How do we know which graph to use: the histogram, stemplot, or dotplot? Since for the most part we are not going to deal with very small data sets in this course, we will generally display the distribution of a quantitative variable using a histogram generated by a statistical software package.

4.4 Module 4 | Numerical Measures

4.4.1 Introduction

The overall pattern of the distribution of a quantitative variable is described by its shape, center, and spread. By inspecting the histogram, we can describe the shape of the distribution, but as we saw, we can only get a rough estimate for the center and spread. A description of the distribution of a quantitative variable must include, in addition to the graphical display, a more precise numerical description of the center and spread of the distribution. In this section we will learn:

- how to quantify the center and spread of a distribution with various numerical measures;
- some of the properties of those numerical measures; and
- how to choose the appropriate numerical measures of center and spread to supplement the histogram.

4.4.2 Module 4 | Measures of Center (1 of 2)

Learning Objectives

- Relate measures of center and spread to the shape of the distribution, and choose the appropriate measures in different contexts.

Intuitively speaking, the numerical measure of center is telling us what is a “typical value” of the distribution.

The three main numerical measures for the center of a distribution are the *mode*, the *mean* and the *median*. Each one of these measures is based on a completely different idea of describing the center of a distribution. We will first present each one of the measures, and then compare their properties.

Mode So far, when we looked at the shape of the distribution, we identified the mode as the value where the distribution has a “peak” and saw examples when distributions have one mode (unimodal distributions) or two modes (bimodal distributions). In other words, so far we identified the mode visually from the histogram.

Technically, the mode is the most commonly occurring value in a distribution. For simple datasets where the frequency of each value is available or easily determined, the value that occurs with the highest frequency is the mode.

Example **(F) Best Actress Oscar Winners**

We will continue with the Best Actress Oscar winners example.

To find the most commonly occurring, or *modal*, age, it is helpful to list the ages in a frequency table, which gives the following results:

Best Actress Age	21	22	25	26	27	28	29	30	31	32	*33*	34	35	36	37	38	39	41	42	44	49	61	62	74	80
Count	1	1	1	2	1	1	3	1	1	2	6	2	2	2	1	1	1	3	2	2	2	2	1	1	1

The mode is 33, since it occurs the most times (6).

Example **(G) World Cup Soccer**

Often, we have large sets of data and use a frequency table to display the data more efficiently.

Data were collected from the last three World Cup soccer tournaments. A total of 192 games were played. The table below lists the number of goals scored per game (not including any goals scored in shootouts).

Total # Goals/Game	Frequency
0	17
1	45
2	51
3	37
4	25
5	11
6	3
7	2
8	1

We can see that the most frequently occurring value is 2 goals (which occurred 51 times). Therefore, the mode for this set of data is 2.

Mean The mean is the average of a set of observations (i.e., the sum of the observations divided by the number of observations). If the n observations are $x_1, x_2, \dots, x_n$, their mean, which we denote by $\bar{x}$ (and read 'x-bar'), is therefore

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}.$$

Example **(H) Best Actress Oscar Winners**

Again we use the Best Actress Oscar winners example.

34 34 27 37 42 41 36 32 41 33 31 74 33 49 38 61 21 41 26 80 42 29 33 36 45 49 39 34 26 25 33 35 35 28 30 29 61 32 33 45 29 62 22 44

The mean age of the 32 actresses is

$$\bar{x} = \frac{34 + 34 + 27 + \dots + 62 + 22 + 44}{44} = \frac{1687}{44} = 38.3.$$

Note that the mean gives a measure of center that is higher than our approximation of the center from looking at the histogram (which was 35). The reason for this will be clear soon.

Example **(I) World Cup Soccer**

We now continue with the data from the last three World Cup soccer tournaments. A total of 192 games were played. The table below lists the number of goals scored per game (not including any goals scored in shootouts).

Total # Goals/Game	Frequency
0	17
1	45
2	51
3	37
4	25
5	11
6	3
7	2
8	1

To find the mean number of goals scored per game, we would need to find the sum of all 192 numbers, then divide that sum by 192. Rather than add 192 numbers, we use the fact that the same numbers appear many times. For example, the number 0 appears 17 times, the number 1 appears 45 times, the number 2 appears 51 times, etc.

If we add up 17 zeros, we get 0. If we add up 45 ones, we get 45. If we add up 51 twos, we get 102. Repeated addition is multiplication.

Thus, the sum of the 192 numbers $= 0(17) + 1(45) + 2(51) + 3(37) + 4(25) + 5(11) + 6(3) + 7(2) + 8(1) = 453$.

The mean is $453/192 = 2.359$.

This way of calculating a mean is sometimes referred to as a *weighted average*, since each value is "weighted" by its frequency. Note that, in this example, the values of 1, 2, and 3 are most heavily weighted.

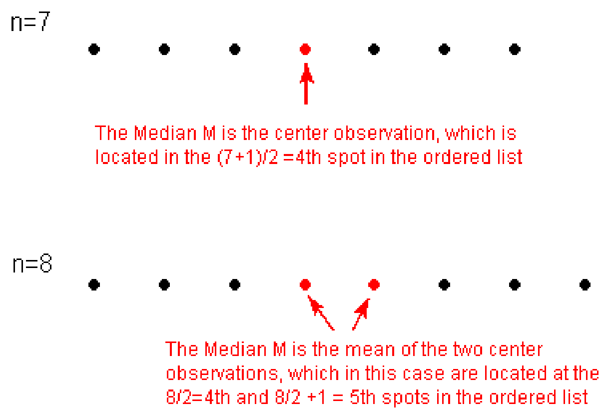
Median The median M is the midpoint of the distribution. It is the number such that half of the observations fall above, and half fall below. To find the median:

- Order the data from smallest to largest.
- Consider whether n, the number of observations, is even or odd.

- If n is *odd*, the median M is the center observation in the ordered list. This observation is the one "sitting" in the *(n + 1)/2 spot* in the ordered list.
- If n is *even*, the median M is the *mean* of the*two center observations* in the ordered list. These two observations are the ones "sitting" in the *n/2* and *n/2 + 1* spots in the ordered list.

Example (J) Median (1)

For a simple visualization of the location of the median, consider the following two simple cases of n = 7 and n = 8 ordered observations, with each observation represented by a solid circle:



Example (K) Median (2)

To find the median age of the Best Actress Oscar winners, we first need to order the data. It would be useful, then, to use the stemplot, a diagram in which the data are already ordered.

Here n = 44 (an even number), so the median M, will be the mean of the two center observations. These are located at the n / 2 = 44 / 2 = *22nd* and n / 2 + 1 = 44 / 2 + 1 = *23rd* spots. Counting from the top, we find that:

- the 22nd ranked observation is 34
- the 23rd ranked observation is 35

Therefore, the median M = (34 + 35)2 = 34.5

```
2|12
2|56678999
3|012233333344
3|5566789
4|1112244
4|599
5|
5|
6|112
6|
7|4
7|
8|0
```

4.4.3 Module 4 | Measures of Center (2 of 2)

Learning Objectives

- Relate measures of center and spread to the shape of the distribution, and choose the appropriate measures in different contexts.

Comparing the Mean and the Median As we have seen, mean and the median, two of the common measures of center, each describe the center of a distribution of values in a different way. The mean describes the center as an average value, in which the *actual values* of the data points play an important role. The median, on the other hand, locates the middle value as the center, and the *order* of the data is the key to finding it.

To get a deeper understanding of the differences between these two measures of center, consider the following example.

Here are two datasets:

Data set A $\longrightarrow$ 64 65 66 68 70 71 *73*

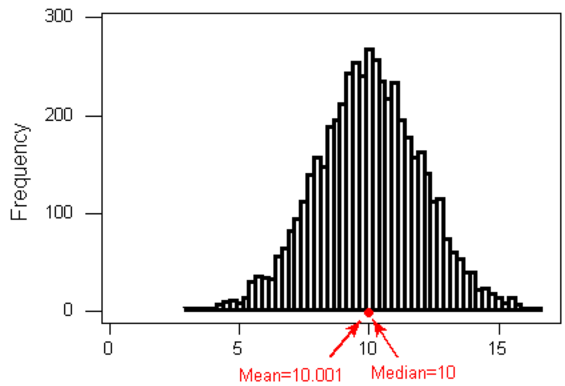
Data set B $\longrightarrow$ 64 65 66 68 70 71 *730*

For dataset A, the mean is 68.1, and the median is 68. Looking at dataset B, notice that all of the observations except the last one are close together. The observation 730 is very large, and is certainly an outlier. In this case, the median is still 68, but the mean will be influenced by the high outlier, and shifted up to 162. The message that we should take from this example is:

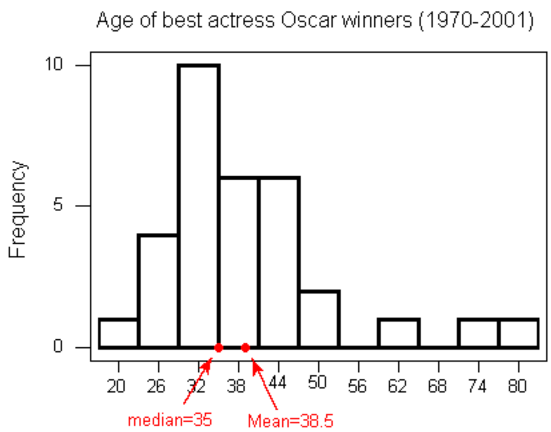
The mean is very sensitive to outliers (because it factors in their magnitude), while the median is resistant to outliers.

Therefore:

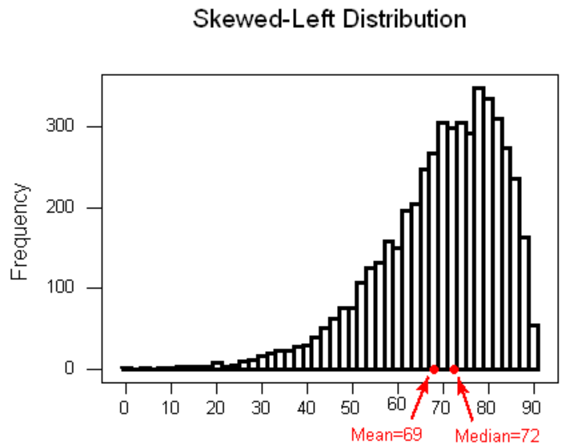
For symmetric distributions with no outliers: $\bar{x}$ is approximately equal to M.



For skewed right distributions and/or datasets with high outliers: $\bar{x} > M$



For skewed left distributions and/or datasets with low outliers: $\bar{x} < M$



We will therefore use $\bar{x}$ as a measure of center for symmetric distributions with no outliers. Otherwise, the median will be a more appropriate measure of the center of our data.

4.4.4 Let's Summarize

- The three main numerical measures for the center of a distribution are the mode, mean $\bar{x}$, and the median (M). The mode is the most frequently occurring value. The mean is the average value, while the median is the middle value.
- The mean is very sensitive to outliers (as it factors in their magnitude), while the median is resistant to outliers.
- The mean is an appropriate measure of center only for symmetric distributions with no outliers. In all other cases, the median should be used to describe the center of the distribution.